SMART ATTENDANCE SYSTEM

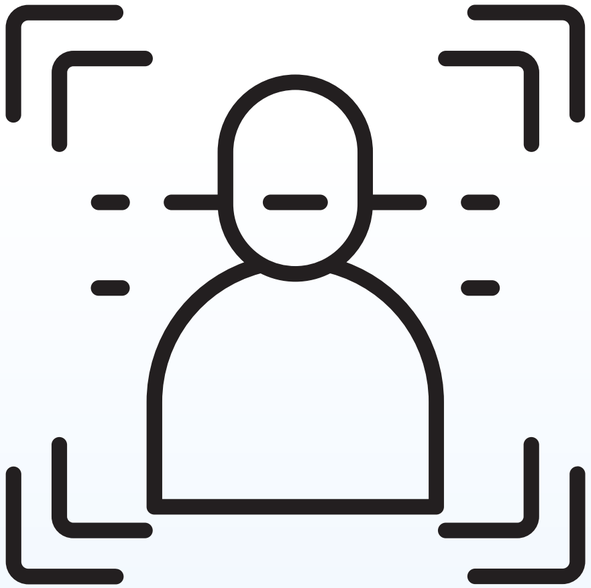
USING FACE RECOGNITION

GROUP MEMBERS:
CHAI YEE PEI
JOEY LOW YAN HUI
TAN JUN YEE
TERENCE LIM CHIA MAO

cheese!

MART ATTENDANCE SYSTE

Smart Attendance Face Recognition System



- A Flutter-based mobile application designed to automate student attendance through facial recognition.
- Lecturers capture a classroom photo via the app, the system identifies registered students using the Face++ API and updates the database in real-time.

—

- Frontend: Flutter (Mobile)
- Backend: Python & MySQL
- Biometrics: Face++ API
- Key Users: Lecturers (Primary users), Super Administrators (System managers), and Students.

Inefficiency

Manual roll calls and sign-in sheets are time-consuming and disrupt valuable teaching time.

Human Error

Traditional records are prone to missing entries or incorrect data logging.

Attendance Fraud (Proxying)

Sign-in sheets: Students can sign for absent peers.
QR Codes: Codes are easily shared via messaging apps with students not physically present.

Hardware Limitations

RFID/Smart card systems require expensive infrastructure and are vulnerable to card swapping or loss.

Regional Gap

Limited adoption of modern biometric solutions in colleges within the northern region of Malaysia.

The Challenges of Conventional Attendance Methods

AIMS AND GOALS

OF THE SMART ATTENDANCE SYSTEM

Automate Identification

To replace manual tracking with an AI-driven facial recognition process to verify student identities.

Enhance Data Integrity

To minimize "proxy attendance" by ensuring only students physically present in the classroom are recorded.

Improve Administrative Efficiency

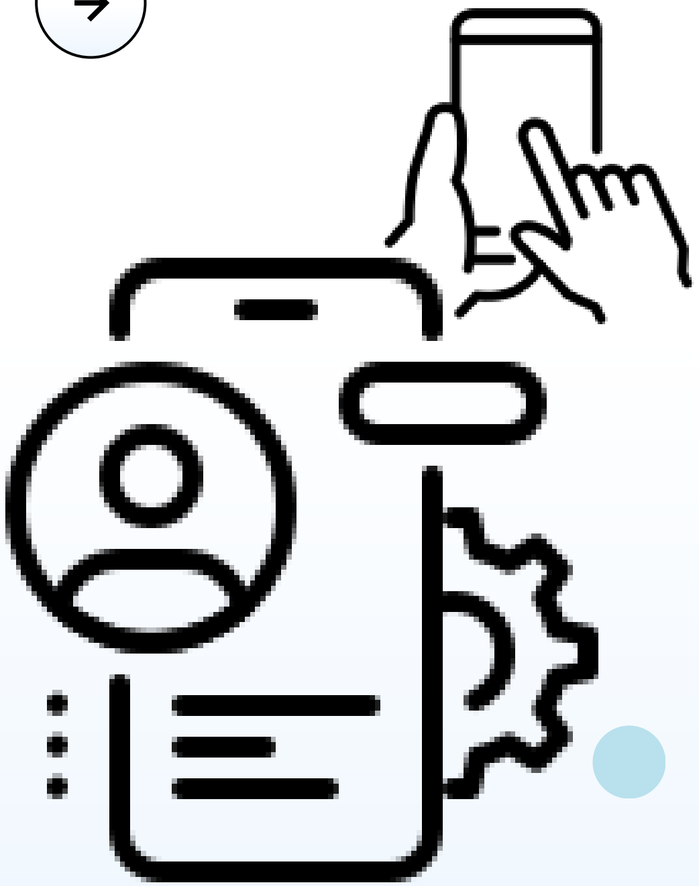
To reduce the workload for teaching staff by automating report generation and record storage.

Digital Transformation

To integrate modern biometric technology (Face++) into the educational environment to support paperless administration.

Ensure Compliance & Security

To develop a system that adheres to PDPA 2010 regulations, ensuring ethical handling of biometric data and user privacy.



Thank you very much!

Presented by Group 5

cheese!
SMART ATTENDANCE SYSTEM